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Chapter-5 (Fiber Joints and couplers)

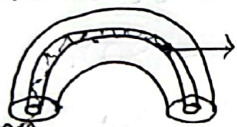
Fiber joints are two types:

1. Fiber splices: these are semi-permanent or permanent joint.

2. Demountable fiber connector or simple connector: these are removable

joints.
* When joint fiber to fiber following losses are generated:

(imp) $\Rightarrow$ Radiative Loss: They are two types:



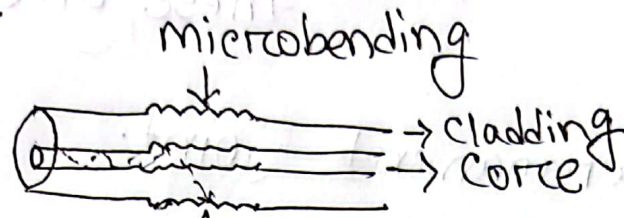
1. Macroscopic Fiber Bending Loss:

Due to greater core radius. Bending in which complete fiber undergoes bends which causes certain mode ^{not} to be reflected and therefore causes loss to the cladding.

2. Microscopic Fiber Bending Loss:

Bending either the core or cladding undergoes slight bends at its

surface. It causes light to be reflected at angles when there is no further reflection.



⇒ Core or Longitudinal misalignment

⇒ Lateral misalignment

⇒ Angular misalignment Losses

⇒ Gap Losses

⇒ Losses due to difference in diameter

⇒ Losses due to change in NA.

* Loss due to misalignment
Mis alignment may occur in three dimensions

1. Longitudinal misalignment $\frac{\circ}{\circ}$ the separation between the fibers.
অনুদৈর্ঘ্য স্থান পরিবর্তন করণ

2. Lateral misalignment $\frac{\circ}{\circ}$ the offset perpendicular to the fiber core axes.
পার্শ্বীয় / আড়াআড়ি ভাবে অবস্থিত
Reduces overlap between two fibers

3. Angular misalignment $\frac{\circ}{\circ}$ the angle between core axes.

* In a multimode step index fiber, the overlapped area between both fiber core approximately gives lateral coupling efficiency η_{lat} .

$$\eta_{lat} = \frac{16 \left(\frac{n_1}{n}\right)^2}{\left(1 + \frac{n_1}{n}\right)^4 \pi} \left\{ \frac{2 \cos^{-1}\left(\frac{y}{2a}\right) - \left(\frac{y}{a}\right) \left[1 - \left(\frac{y}{2a}\right)^2 \right]^{\frac{1}{2}}}{\cos^{-1}\left(\frac{y}{2a}\right) \times \frac{\pi}{180}} \right\}$$

y = lateral offset / of the fiber
 a = fiber core radius.

imp * A step index fiber has a core refractive index of 1.5 and a core diameter of 50 μm . The fiber is joint with lateral misalignment between the core axis 5 μm . Estimate insertion loss at the joint due to the lateral misalignment assuming a uniform distribution of power between all guided modes when:

- there is a small air gap.
- the joint is considered index matched.

Solution:

a) Given, refractive index, $n_1 = 1.5$
as there is small air gap,
 $n = 1$

Diameter = 50 μm

radius, $a = \frac{50}{2} = 25 \mu\text{m}$

lateral misalignment, $y = 5 \mu\text{m}$

We know,

$$n_{lat} = \frac{16 \left(\frac{n_1}{n}\right)^2}{1 + \left(\frac{n_1}{n}\right)^4} \frac{1}{\pi} \left\{ 2 \cos^{-1} \left(\frac{y}{2a} \right) - \left(\frac{y}{a} \right) \left[1 - \left(\frac{y}{2a} \right)^2 \right]^{\frac{1}{2}} \right\}$$

$$= \frac{16 \times \left(\frac{1.5}{1}\right)^2}{\left(1 + \left(\frac{1.5}{1}\right)^4\right)} \times \frac{1}{\pi} \times \left\{ 2 \cos^{-1} \left(\frac{5}{2 \times 25} \right) - \left(\frac{5}{25} \right) \left[1 - \left(\frac{5}{2 \times 25} \right)^2 \right]^{\frac{1}{2}} \right\}$$

$$= 0.29335 \times 2 \cos^{-1} \left(\frac{5}{50} \right) - 0.198998$$

$$= 0.293 \times \left\{ 2.941 - 0.2 (0.99)^{\frac{1}{2}} \right\}$$

$$= 49.377 - 0.199 = 49.178$$

$$= 49.178$$

The insertion loss due to lateral misalignment

$$LOS_{lat} = -10 \log_{10} n_{lat} \text{ dB}$$

$$= -10 \log_{10} (49.178)$$

$$= -16.918$$

$$= 0.948 \text{ dB}$$

b) When the joint is considered, in matched (no air gap) the coupling efficiency may again obtain,

$$\eta_{lat} \approx \frac{1}{\pi} \left\{ 2 \cos^{-1} \left(\frac{5}{50} \right) - \left(\frac{5}{25} \right) \left[1 - \left(\frac{5}{50} \right)^2 \right]^{\frac{1}{2}} \right\}$$

$$\approx \frac{1}{\pi} \left\{ 2 \times 1.471 - 0.2 (0.99)^{\frac{1}{2}} \right\}$$

$$\approx 0.872$$

insertion

Therefore, loss

$$\text{Los}_{lat} = -10 \log_{10} (0.872)$$

$$\text{Los}_{lat} = 0.59 \text{ dB}$$

With index matching the insertion loss at the joint is reduced

$$= 0.95 - 0.59$$

$$= 0.36 \text{ dB}$$

* Lateral misalignment loss was dependant on the refractive index gradient $\alpha=2$

$$L_t = \frac{2}{\pi} \left(\frac{y}{a} \right) \left(\frac{\alpha+2}{\alpha+1} \right) \text{ for } 0 \leq y \leq 0.2a$$

Lateral coupling efficiency,

$$\eta_{lat} = 1 - L_t$$

* A graded index fiber has a parabolic refractive index profile ($\alpha=2$) and core diameter of $50 \mu m$. Estimate insertion loss due to a $3 \mu m$ lateral misalignment at a fiber joint when there is index matching and assuming.

a) there is uniform illumination of all guided modes only.

b) there is uniform illumination of all guided and leaky

for cases of guided plus leaky modes.
 " guided only
 " both guided plus leaky modes.
 where $\alpha=2$, $L_t = 0.75 \left(\frac{y}{a} \right)$
 where, $\alpha=80$, $L_t = 0.6 \left(\frac{y}{a} \right)$
 " $L_t = 0.5 \left(\frac{y}{a} \right)$
 " $L_t = 0.5 \left(\frac{y}{a} \right)$
 ** all guided and leaky

a) Assuming uniform illumination of guided modes only.

$$\alpha = 2$$

$$y = 3 \mu\text{m}$$

$$\text{diameter } r = 50 \mu\text{m}$$

$$a = \frac{50}{2} = 25 \mu\text{m}$$

we know,

$$L_t = \frac{2}{\pi} \left(\frac{y}{a} \right) \left(\frac{\alpha + 2}{\alpha + 1} \right)$$

$$= \frac{2 \times 3 \times 4}{\pi \times 25 \times 3}$$

$$= 0.102$$

The coupling efficiency,

$$\eta_{\text{lat}} = 1 - L_t = 1 - 0.102$$

$$= 0.898$$

Hence, insertion loss,

$$\text{Loss}_{\text{lat}} = -10 \log_{10} \eta_{\text{lat}}$$

$$= -10 \log_{10} (0.898)$$

$$= 0.47 \text{ dB}$$

b) Assuming the uniform illumination of both guided and leaky mode,

$$L_t = 0.75 \left(\frac{a}{a} \right) = 0.75 \left(\frac{3}{25} \right)$$

$$= 0.09$$

∴ Lateral coupling efficiency,

$$\eta_{lat} = 1 - L_t = 1 - 0.09 = 0.91$$

∴ insertion loss,

$$Loss_{lat} = -10 \log(0.91) = 0.41 \text{ dB}$$

* Angular coupling efficiency, $\theta \times \frac{\pi}{180} n$

$$\eta_{ang} \approx \frac{16 \left(\frac{n_1}{n}\right)^2}{\left(1 + \frac{n_1}{n}\right)^4} \left[1 - \frac{n \theta}{\pi n_1 (2\Delta)^{\frac{1}{2}}} \right]$$

θ = angular displacement

Δ = relative refractive index difference

* Insertion Loss due to angular misalignment

$$\text{Loss}_{ang} = -10 \log_{10} \eta_{ang}$$

Imp * Two multimode step index fibers have numerical aperture of 0.2 and 0.4 respectively, and both have the same core refractive index of 1.48. Estimate the insertion loss at a joint in each fiber caused by a 5° angular misalignment of the fiber core axis. It may be assumed that the medium between the fibers is air.

Solution:

i. The angular

Given that, $NA_1 = 0.2$

$$NA_2 = 0.4$$

$$n_1 = 1.48$$

$$\theta = 5^\circ$$

$$n = 1$$

We know,

The angular coupling efficiency,

$$\eta_{\text{ang}} = \frac{16 \left(\frac{n_1}{n}\right)^2}{\left(1 + \frac{n_1}{n}\right)^4} \left[1 - \frac{n\theta}{\pi n_1 (2\Delta)^{\frac{1}{2}}} \right]$$

$$= \frac{16 \cdot (1.48)^2}{(1 + 1.48)^4} \left[1 - \frac{5}{\pi \times \frac{1.48}{n_1} \times (2\Delta)^{\frac{1}{2}}} \right]$$

For $NA_1 = 0.2$,

$$NA \approx n_1 (2\Delta)^{\frac{1}{2}}$$

$$\therefore \eta_{\text{ang}} = 0.9265 \left[1 - \frac{5 \times \frac{\pi}{180}}{\pi \times NA} \right]$$
$$= 0.9265 \left[1 - \frac{5 \times \frac{\pi}{180}}{\pi \times 0.2} \right]$$

$$= 0.7978$$

∴ Insertion Loss,

$$\text{Loss}_{\text{ang}} = -10 \log_{10} \eta_{\text{ang}} = -10 \log_{10} (0.7978)$$

$$= 0.98 \text{ dB (Ans)}$$

when, $NA_2 = 0.4$

$$\eta_{\text{ang}} = 0.9265 \times \left[1 - \frac{5 \times \frac{\pi}{180}}{\pi \times 0.4} \right]$$

$$= 0.862$$

∴ Insertion Loss,

$$\text{Loss}_{\text{ang}} = -10 \log_{10} (0.862)$$

$$= 0.649 \text{ dB (Ans)}$$

*Fiber splices :

Permanent and semi-

permanent joint between two optical fiber

⇒ Fusion splice

⇒ Mechanical splice

⇒ Precision tube splice

⇒ Loose tube splice

⇒ V-groove splice

⇒ Precision pin splice

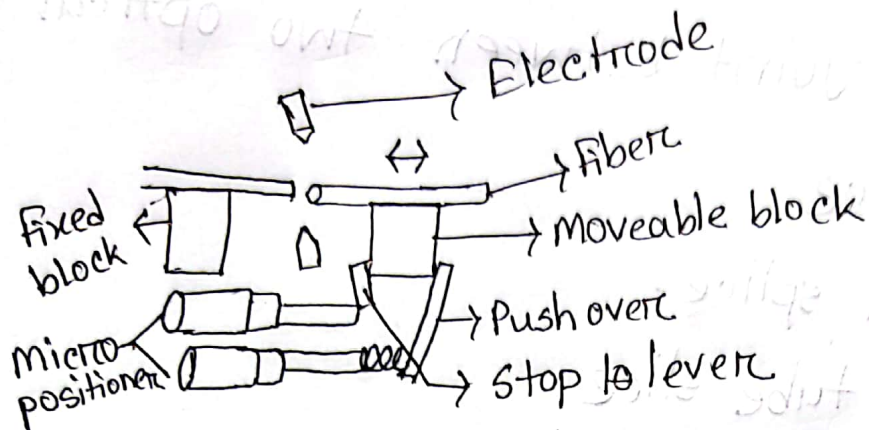
⇒ Multiple splice.

*Fusion splice: involves the heating of the

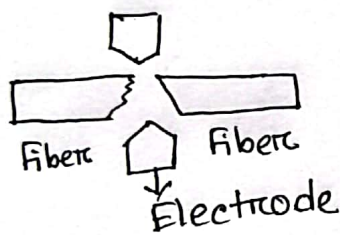
two prepared fiber ends to their fusing point with the application of sufficient axial pressure between the two optical fibers. The stripped fiber ends are adequately positioned and aligned

in order to achieve good continuity of transmission medium at the joint junction point. Fibers usually positioned and clamped

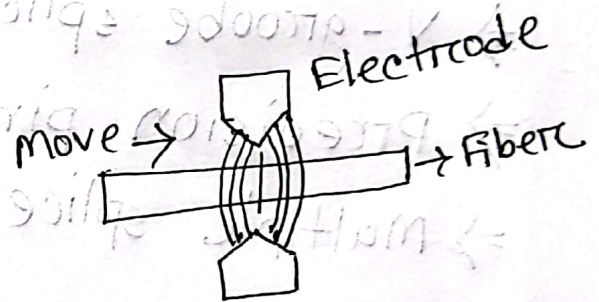
with the aid of inspection microscope.
Heating source is an electric arc.



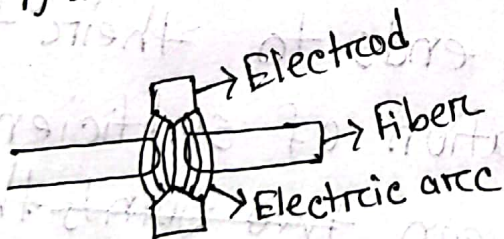
a) An example of fusion splicing apparatus.



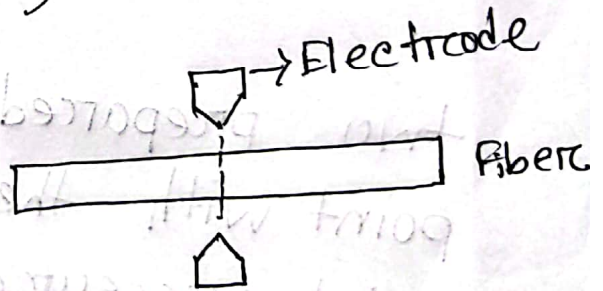
1) Initial setting



3) Pressed together



2) arrangement of smooth surface by pre-fusion

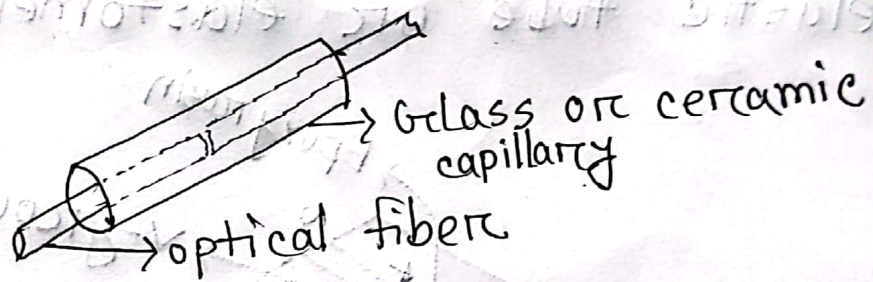


4) Accomplishment of splice.

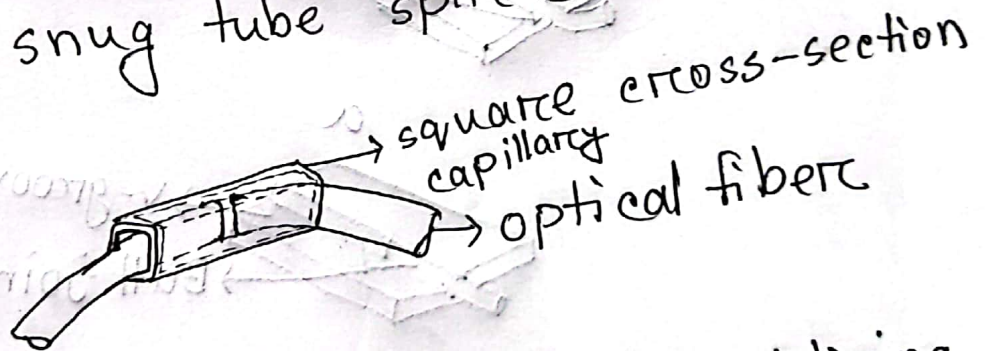
b) schematic illustration of pre-fusion splicing optical method for accurately splicing fiber

ope. *** Mechanical splice ***

utilize a glass or ceramic with an inner diameter just large enough to accept the optical fibers. Transparent adhesive is injected to give mechanical sealing and index matching of the splice.



a) snug tube splice



b) loose tube splice utilizing square cross-section capillary

Fig! Technique for tube splicing of optical fiber

*V-groove splice! Mechanical splicing

involves the use of grooves to secure the fibers to be joined. The splice is made permanent by securing the fibers in the V-groove with epoxy resin. The V-groove technique is the elastic tube or elastomeric splice.

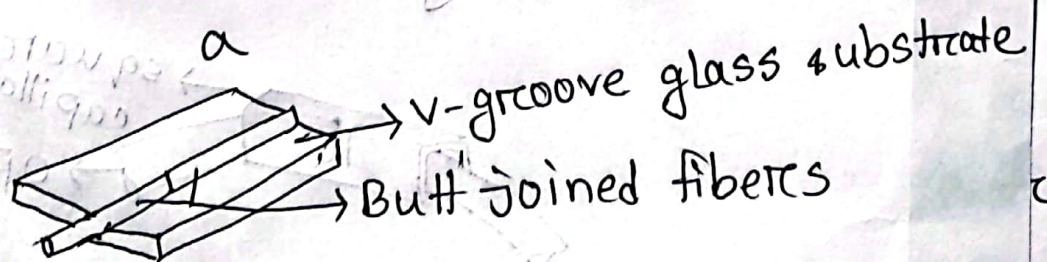
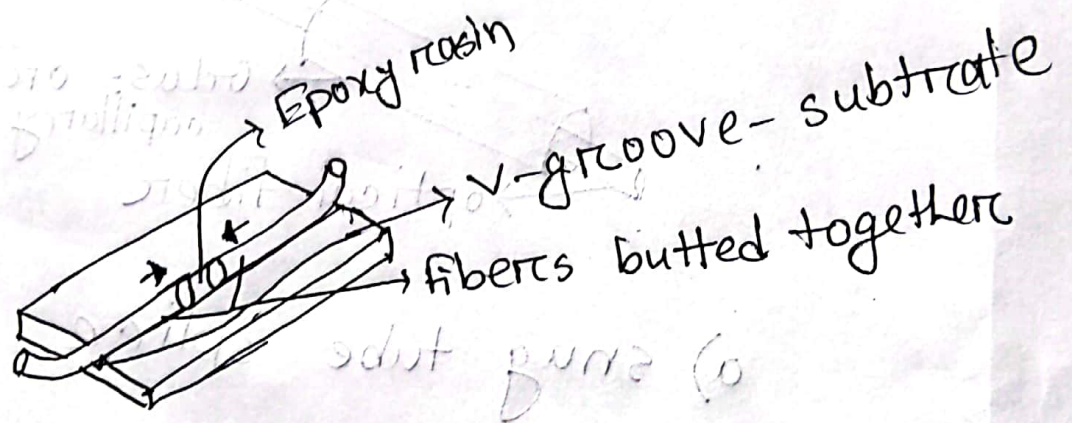


Fig: V-groove splices.

*Multiple splices:

Multiple simultaneous fusion splicing of an array of fibers in a ribbon cable has been demonstrated for both multimode and single mode - fibers.

*Fiber couplers:

Fiber couplers are branching devices that split all the light from a main fiber into two or more fibers.

18-19 → 1(c)
* "Longitudinal misalignment gives significantly minor losses per unit displacement than Lateral misalignment" — True or False. Justify the statement.

Ans:

True. Longitudinal misalignment only affects the axial load slightly, causing minimal disruption to the smooth transfer of rotational power. Lateral misalignment creates greater stress on components by inducing significant radial forces.

and vibrations which disrupts the power transmission and results higher energy loss. Lateral misalignment can lead to increased friction which causes greater losses per unit displacement. So, Longitudinal is less damaging and have minor loss per unit displacement than Lateral.

(15-10-21)

There are three types of misalignment: longitudinal, lateral and angular. Longitudinal misalignment occurs when the shafts are not perfectly aligned along their length. Lateral misalignment occurs when the shafts are not perfectly aligned in the horizontal plane. Angular misalignment occurs when the shafts are not perfectly aligned in the vertical plane. Longitudinal misalignment causes the shafts to rub against each other, which increases friction and heat. Lateral misalignment causes the shafts to bend, which also increases friction and heat. Angular misalignment causes the shafts to twist, which can lead to fatigue and failure.